Have you noticed that, so far, we only have drawn graphs using segments? While this has so-far served our purpose well, and we were always able to draw graphs with the minimum number of intersections, in general this is not always true.

In fact, we need to introduce a new definition: the **`rectilinear crossing number`** of a graph `G` is the minimum number of edge crossing in a straight-line drawing of `G`, i.e. an embedding on the plane for `G`, where edges are drawn as straight-line segments.

When we restrict to straight-line segments for the edges, we are using a fraction of the possible embeddings; it shouldn't be surprising, therefore, that the rectilinear crossing number of a graph is never smaller than its crossing number.

Now, can it be larger? It has been proved that, for any graph `G` with a crossing number smaller or equal than 3, it's possible to come up with a straight-line drawing with minimal crossings: in other words, whenever the crossing number `cr(G)` is 3 or less, it matches the rectilinear crossing number, `rcr(G)`.

